

Four-Style Video Captioner

Gemma watches the video. Gemma writes the caption.

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AMD Developer Hackathon Act II · Track 2

The task, and what actually scores

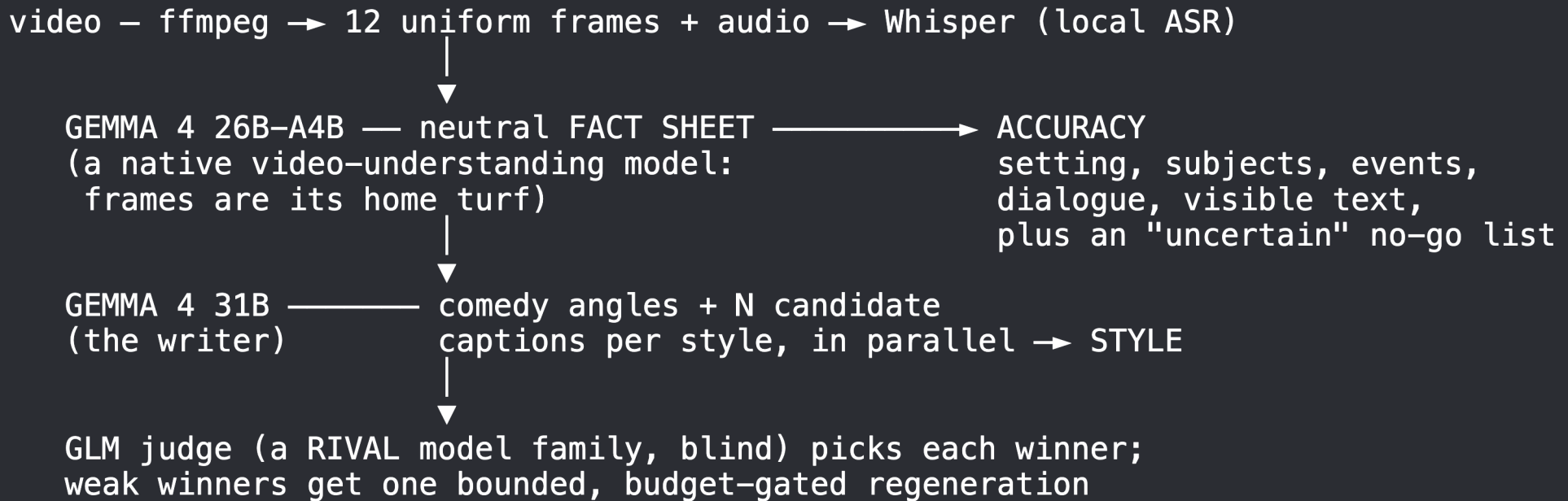
One clip in, four captions out: **formal, sarcastic, humorous_tech, humorous_non_tech**.

An LLM judge scores every caption twice:

- **Accuracy**: does the caption reflect what is actually in the video?
- **Style**: is the tone unmistakable?

Asking one model to "watch a video and be funny" trades facts for jokes. So we never ask that.

Architecture: perception and style, separated



Gemma is the core, not a garnish

- **Every fact and every caption is Gemma's work.** Gemma 4 26B-A4B reads the frames; Gemma 4 31B finds the comedy and writes all four styles.
- The judge is deliberately **not** Gemma: a rival family referees Gemma's captions blind. Stronger evidence than Gemma grading itself.
- Served via the Gemini API, with a Gemma-on-OpenRouter fallback tier, then Fireworks as last resort. **Every fallback is logged** to `run_report.json` next to the results, so the Gemma claim is auditable, not aspirational.
- The config reserved these stages for Gemma from day one.

Grounding beats hallucination

- The fact sheet carries an **"uncertain" list**: anything the vision pass could not verify is a no-go for every later stage.
- Comedy angles must cite the fact sheet. The writer must anchor each caption in a concrete, verified detail.
- **humorous_non_tech is hard-guarded**: a deterministic vocabulary check (with plural and punctuation dodges covered) rejects any caption with tech jargon before it can ship.
- Whisper output is filtered for the classic "Thanks for watching" style hallucinations before it can enter the facts.

Reliability engineering (a timeout scores zero)

- **Watchdog**: a valid `results.json` is force-flushed before the 10-minute wall, no matter what a network call is doing.
- **Per-clip deadlines** with a degradation ladder: skip ASR, then skip comedy, then single-candidate fast path — a slow clip gets a plainer caption, never a zero.
- **No caption is ever empty**: judge failure falls back to the best candidate, pipeline failure to a deterministic caption built from the fact sheet, and a final backfill guarantees every requested style ships text.
- Downloads capped by wall-clock and size; all four styles generate in parallel.

Real output (example clip: city traffic, autumn boulevard)

- **formal**: "Heavy traffic flows along a multi-lane avenue in a modern East Asian city, with motion blur indicating continuous movement."
- **sarcastic**: "Three minutes of cinematic motion blur so traffic can feel urgent about remaining exactly the same."
- **humorous_tech**: "Aggressive motion blur to render traffic that literally just continues — peak overengineering."
- **humorous_non_tech**: "The blue bus got one second of fame before the blur ate it, and I have never related to a vehicle more."

Validated end to end

- Full harness contract: reads `/input/tasks.json`, downloads each `video_url`, writes `/output/results.json` with exact style keys.
- Three example clips: **all four styles filled on every clip**, correct keys, ~54s per clip with four-way parallelism — comfortable inside the budget at the hidden set's scale.
- Container: linux/amd64, ffmpeg + Whisper baked in, well under the size limit.

Four-Style Video Captioner

Gemma perceives. Gemma writes. A rival model referees.

github.com/codechirag5610/ai-video-captioner

Thank you